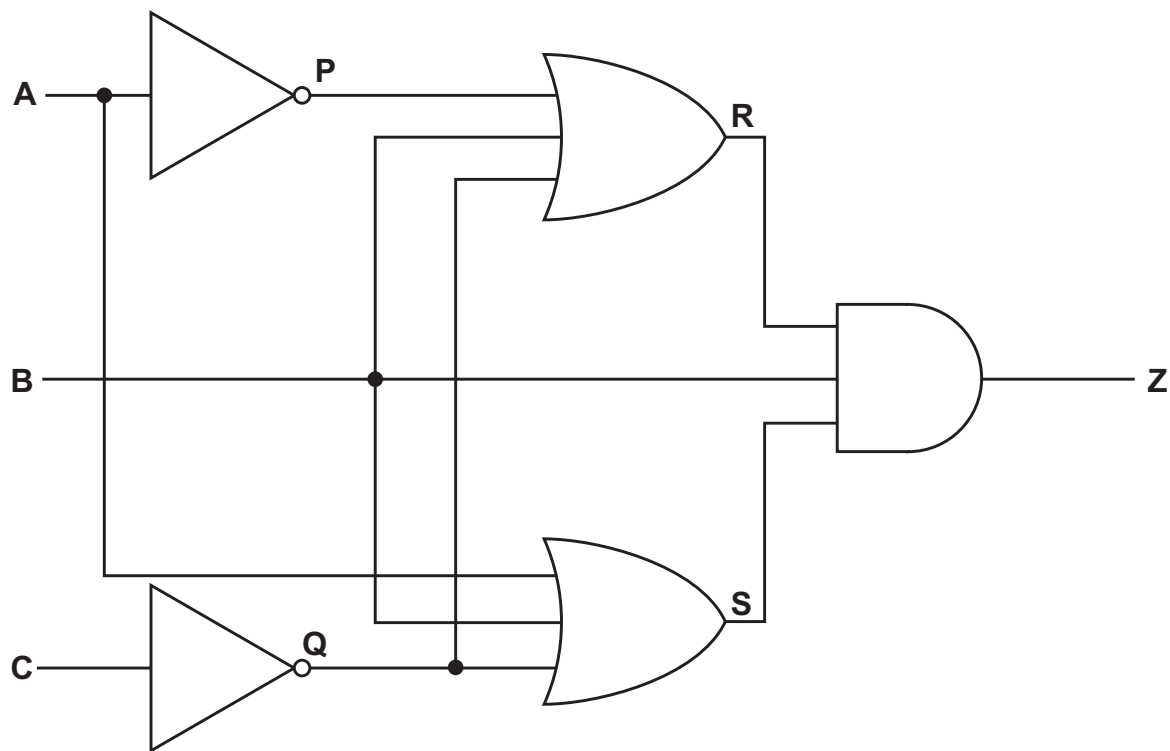


6 (a) The diagram shows a logic circuit.



Complete the truth table for the given logic circuit.
Show your working.

			Working space				
A	B	C	P	Q	R	S	Z
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) (i) Complete the Karnaugh map (K-map) for the Boolean expression:

$\overline{A}.B.C + \overline{A}.B.\overline{C} + A.\overline{B}.C + A.B.\overline{C}$

		BC			
A		00	01	11	10
	0				
	1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products.

[2]

(iii) Write the Boolean expression from your answer to part b(ii) as a simplified sum-of-products. Do **not** carry out any further simplification.

.....

..... [2]